Measurement systems

R0005E - Measurement and Feedback Control

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Terminology in measurement systems

Measurements

- Measurement is the science of determining a value for a physical variable.
- It is the process of obtaining information regarding the value of a physical variable.
- Deriving the quantity of a variable is the result of a comparison between the quantity and a predetermined or predefined standard.
- The quantity can be expressed in numerical values.

Instrumentation

- Device used in measurement systems
- Test instrumentation are devices used during development and approval of engineering systems

Calibration

- For the complete measurement system a relationship to an approved national or international standard is established.

Historical measurement systems

Early measurement systems used body measures or seeds as standard to compare with:

- In Egypt, Indus valley and Mesopotania examples are found
- Right: Measurement tools that were found in the tomb of Senedjem. Plomb, Square, square level.

Example Carat:

- The Carob seeds where used to standardize carat.
- 4 seeds equaled 1 carat.
- Today carat is standardized to 0.2 grams.

Transducer

Obviously, we do convert a quantities between different domains.

A **transducer** is a device that converts one form of energy to another. Typical signal domains are:

- Magnetic
- Mechanical
- Thermal
- Optical
- Chemical
- Electrical

Some special types of transducers:

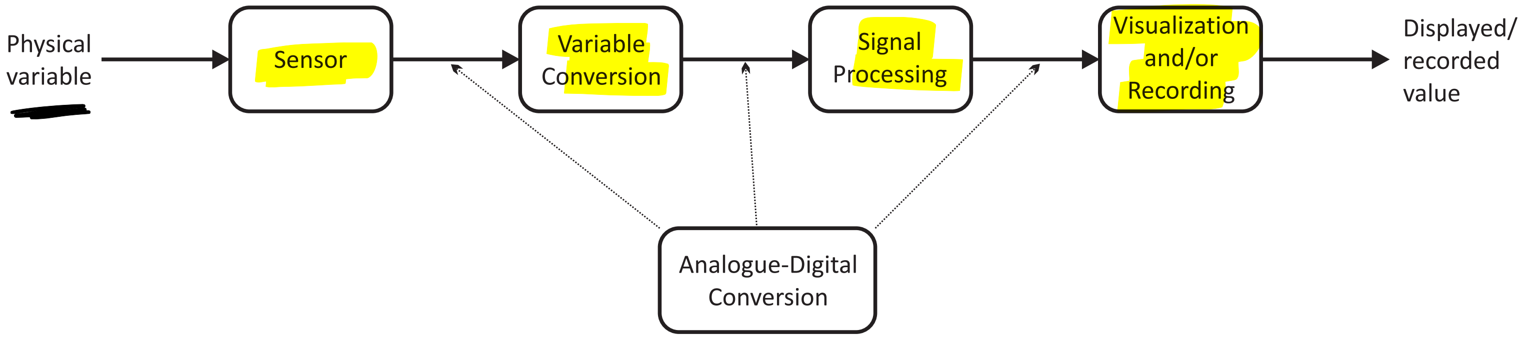
Electronic transducer: Has an input or output that is from the electrical domain, like e.g. voltage, current, resistance.

Sensor: Electronic transducer that converts a physical quantity into an electrical signal. Often digital such that it can be easily processed, transmitted and stored.

Actuator: Electronic transducer that converts an electrical signal into another form of energy. This is quite often mechanical.

Components of a measurement system and Instrument types

A measurement system is composed of the following



Types of instruments:

- **Passive instruments:** The output of the instrument is entirely produced by the quantity being measured.
- **Active instruments:** The quantity being measured modulates the magnitude of some external power source.
- **Analogue instruments:** The output varies continuously as the quantity is being measured.
- **Digital instruments:** The output varies in discrete steps and only has a finite number of values (quantized).
- **Smart instruments:** Equipped with a microprocessor to perform operations before and after the measurements

Static characteristics

Definition:

The static characteristics of an instrument is the steady state relationship between the input and the output of an instrument, and does not involve differential equations.

Important parameters:

- Accuracy and Precision
- Repeatability and reproducibility
- Tolerance
- Range or span
- Linearity
- Sensitivity to measurement and disturbance
- Threshold
- Resolution
- Hysteresis effects
- Dead zone

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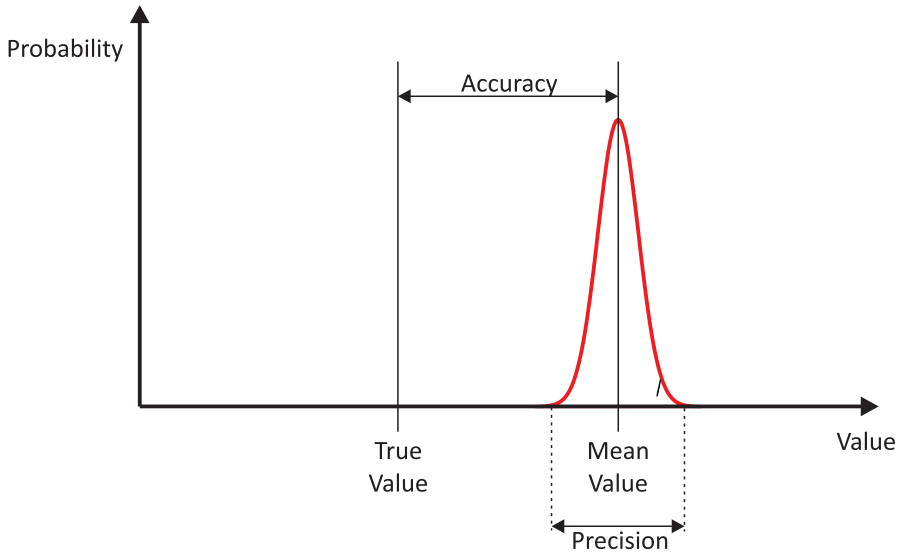
Accuracy and Precision

Accuracy is a measure for an instrument of how close the output of the instrument is to the true value.

Precision is a measure for an instrument of how close recurring measurements are to each other.



Repeatability, Reproducibility, and Tolerance

Repeatability describes the ~~closeness~~ of outputs when the same input is applied repetitively over a short period of time, with the same measurement conditions, same instrument and observer, same location and same conditions of use maintained throughout the period.

Reproducibility describes the closeness of outputs for the same input when there are changes in the method of measurement, observer, measuring instrument, location, conditions of use and time of measurement.

Tolerance describes the maximum error that can be expected in some value, which is the deviation between the true value and the output.

Note:

- Both terms thus describe the spread of output readings for the same input, which relates strongly to precision.
- The precision is referred to as repeatability if the measurement conditions are constant and as reproducibility if the measurement conditions vary.
- Tolerance relates strongly to accuracy.

Range, Resolution, and Threshold

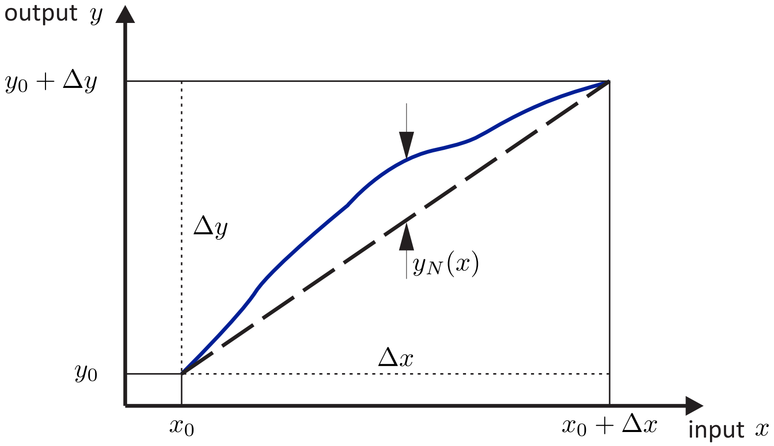
Range is given by the minimum and maximum values of a quanttity that the instrument is designed to measure.

Resolution is the lower limit on the magnitude of the change in the input measured quantity that produces an observable change in the instrument output.

Threshold: If the input to an instrument is gradually increased from zero, the input will have to reach a certain minimum level before the change in the instrument output reading is of a large enough magnitude to be detectable. This minimum level of input is known as the threshold of the instrument.

Linearity and Sensitivity

Linearity is a highly desirable property of an instrument, which means that the output is proportional to the change in the input. The deviation from true linearity is called linearity error.



From the figure we can derive a curve that is composed of a linear and non-linear part

$$y(x) = y_0 + \frac{\Delta y}{\Delta x}(x - x_0) + y_N(x)$$

The sensitivity of the measurement instrument is defined as

$$s(x) = \frac{dy}{dx} = \frac{\Delta y}{\Delta x} + \frac{dy_N(x)}{dx}$$

For an instrument with $y_N(x) = 0$, true linearity is achieved, and the sensitivity becomes

$$s(x) = \frac{\Delta y}{\Delta x}$$

Sensitivity to disturbances

Ambient conditions are never the same and do certainly vary, like for example the ambient temperature or air pressure.

Certain static instrument characteristics may change, and the sensitivity to disturbance is a measure of the magnitude of this change. Such changes in ambient conditions affect instruments in two main ways:

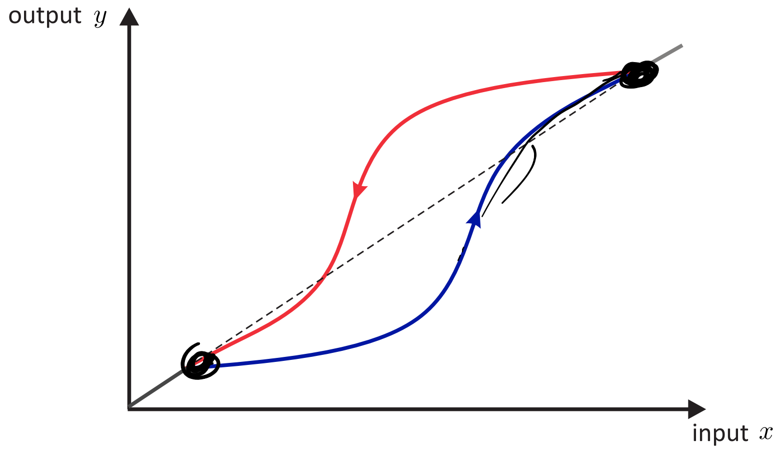
- zero drift or bias.
- sensitivity drift or scale factor drift.

Zero drift or bias describes the effect where the zero reading of an instrument is modified by a change in ambient conditions.

Sensitivity drift describes the amount by which an the sensitivity of an instrument varies as ambient conditions change.

Hysteresis

When the direction of the evolution of the input values matters for the output of the instrument, then there is a hysteresis effect present:



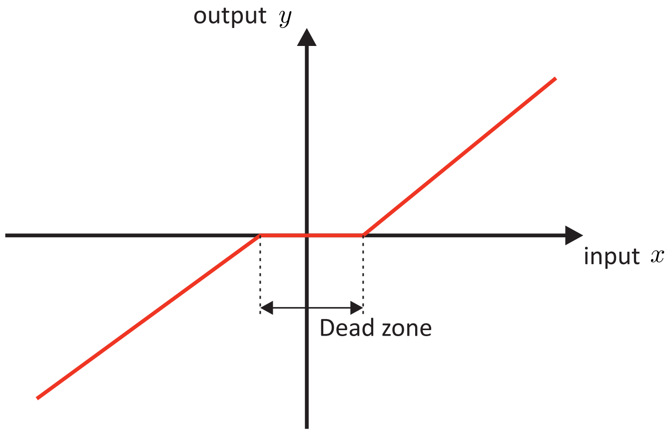
The dashed line expresses the desired linear curve of the instrument, which is distorted by the hysteresis.

Commonly, mechanical instruments, like e.g. the fly-ball and instruments with springs, exhibits hysteresis effects.

Note: Hysteresis is a highly non-linear effect which is usually difficult to deal with.

Dead space or dead zone

Dead space or dead zone is defined as the range of different input values over which there is no change in the output value.



Apparently, instruments that exhibit hysteresis, have a threshold or poor resolution may have dead spaces or dead zones.

Dynamic characteristics

Usually an instrument is build from components that exhibits also dynamic behaviour. In that case, the characteristics of the instrument is governed by a differential equation.

Essentially, the instrument can be described by the use of a transfer function and the prior lectures on models and the transfer functions summarize the characteristics.

Note: The lectures relating to frequency response and the filters also apply for measurement systems and instruments.

Measurement errors

There are two categories of measurement errors and they are defined as follows:

Systematic errors describe errors in the output readings of a measurement system that are consistently on one side of the correct reading, i.e. either all the errors are positive or they are all negative.

Random errors are perturbations of the measurement either side of the true value caused by random and unpredictable effects, such that positive errors and negative errors occur in approximately equal numbers for a series of measurements made of the same quantity. Such perturbations are mainly small, but large perturbations occur from time to time, again unpredictably.

The causes for systematic errors are often disturbances during the measurement process or the change of environmental conditions that have an effect on the measurement. Such errors can often be compensated for.

Quantification of the systematics errors is done by first compensating for them, and thereafter expressing the remaining errors as $\pm x\%$ of the output, assuming a mid-point for the conditions.

On the contrary random errors are stochastic in nature and can only be described by the use of probabilistic distributions. In order to characterize these distributions large number of samples are needed.

Summary of some statistical properties

For a population x of output readings with n members the following is known

Mean:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Variance:

$$\sigma^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} x_i\right)^2\right]$$

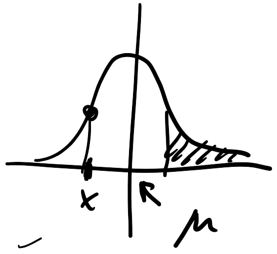
Standard error of the mean: For a finite population versus an infinite population

$$\alpha = \frac{\sigma}{\sqrt{n}}, \quad \bar{x} = \mu \pm \alpha$$

Normal distribution: Probability for a certain reading

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

Gauss -Bell curve



Variance in terms of the distribution

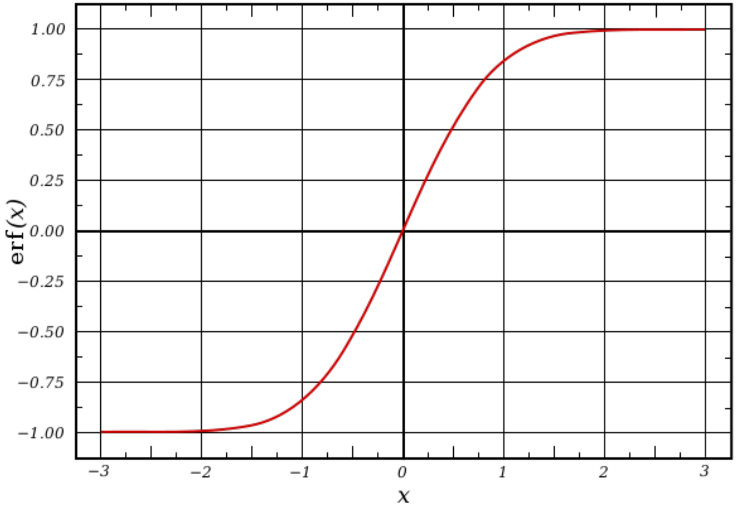
$$\sigma^2 = \int_{-\infty}^{\infty}(x-\mu)^2 f(x)dx$$

Probability for a measurement in an intervall

We make use of the error function to define the cumulative distribution function of the normal distribution.

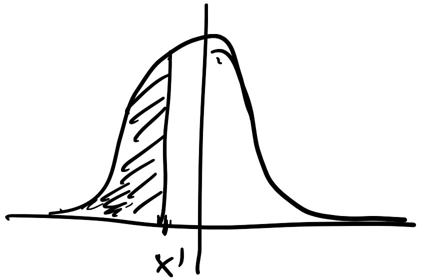
Error function

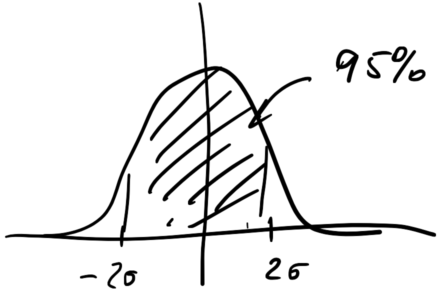
$$erf(x) = \frac{2}{\sqrt{\pi}} \int_0^x exp(-t^2)dt$$



Cumulative distribution function

$$F(x) = \frac{1}{2}\left(1 + erf\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right)$$





Estimation of the random error in a single measurement

For a 95% confidence interval, the maximum likely deviation in a single measurement is $\pm 1.96\sigma$.

However, this only expresses the maximum likely deviation of the measurement from the calculated mean of the reference measurement set, which is not the true value as observed earlier.

Thus the calculated value for the standard error of the mean α has to be added to the likely maximum deviation value. Thus, the maximum likely error in a single measurement can be expressed as:

$$Error = \pm(1.96\sigma + \alpha)$$

Calibration

Calibration consists of comparing

- the output of the instrument or sensor under test against
- the output of an instrument of known accuracy

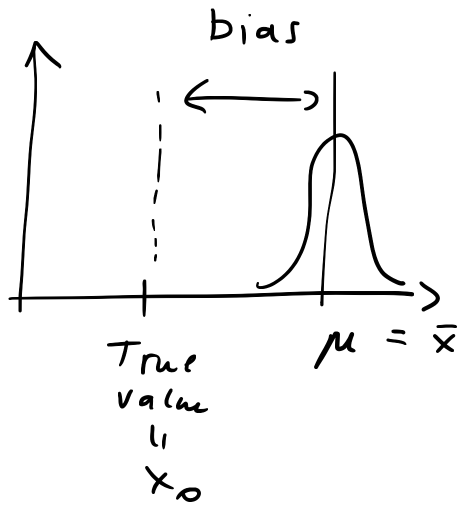
In this process the same input (measured quantity) is applied to both instruments.

Calibration frequency:

- If an instrument is required to measure some quantity and an inaccuracy of $\pm x\%$ is acceptable, then a certain amount of performance degradation can be allowed if its inaccuracy immediately after recalibration is less than $\pm x\%$.
- It is important that the pattern of performance degradation be quantified, such that the instrument can be recalibrated before its accuracy has reduced to the limit defined by the application.

Note:

- The standard instrument for calibration is usually calibrated by an authority.
- During the calibration the environmental conditions need to be controlled and noted.



$$\Delta Y = \underbrace{\bar{x} - x_0}_{\Delta x}$$